Copies, Conflicts and Broken Parses: How Corpus Defects Affect Retrieval-Augmented Claim Verification

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Abstract

A retrieval-augmented system can only be as reliable as the collection it searches, yet the effect of individual collection defects is seldom measured on its own. We inject four common defects into the SciFact corpus, namely near-duplicate copies, stale copies with reversed findings, broken parsing and fixed-size chunking, at rates of 10 to 50% of the gold evidence. We then track, claim by claim, how retrieval and the verdicts of four verifiers change. The verifiers are two zero-shot LLMs (Qwen2.5 1.5B and 3B) and an NLI cross-encoder, used either off the shelf or fine-tuned. Every comparison is repeated over five corruption seeds and three training seeds, and the main experiments are replicated on Climate-FEVER. Copies push the gold document out of first place (dense recall@1 falls from 0.74 to 0.53 when half of the gold abstracts have a stale copy), but they hardly change the final verdicts. Broken parsing behaves the other way round. Retrieval barely registers it, yet it costs correct answers for both LLMs and for the fine-tuned verifier under every seed. Stale copies pull the fine-tuned verifier towards CONTRADICT; on the one-sentence records of Climate-FEVER they cost it up to 55 correct answers for 14 gained and make it more often confidently wrong, while both LLMs largely ignore them. Single-sentence chunks make the fine-tuned verifier both less accurate and more confident. A team that monitors retrieval quality alone would therefore rank these defects in the wrong order of harm.

1 Introduction

Retrieval-augmented generation (RAG; Lewis et al., 2020) answers a question from documents it retrieves out of a collection, so the answer can be no better than what the collection contains. Real collections are rarely clean. The same document arrives from several sources, an outdated version sits next to the revision that contradicts it, PDF extraction truncates or scrambles text, and chunking cuts documents into pieces whose size is often chosen by habit. Most robustness studies of RAG intervene on the reader's context, for example by inserting irrelevant or counterfactual passages (Cuconasu et al., 2024; Chen et al., 2024; Xie et al., 2024). In practice, though, engineers do not edit the context passage by passage. They maintain a collection, and they need to know which of its problems are worth the cost of cleaning.

This paper asks how much each of four common defects, taken on its own, changes retrieval and the final answer. We study claim verification because its datasets label both the verdict (does the evidence support or contradict the claim?) and the evidence itself, which lets us follow a defect from the retriever to the verdict. We inject one defect type at one rate at a time into an otherwise untouched corpus and keep every model, prompt and threshold fixed, so that any change in the results can be attributed to the corpus.

We make three contributions. First, we release a seeded corruption tool for four defect types that logs every change it makes, together with a pipeline that regenerates each corrupted corpus from its seed and checks it against the logged changes. Second, we report paired, claim-level measurements on SciFact (Wadden et al., 2020) for two retrievers, a reranker and four verifiers, repeated over five corruption seeds and three verifier training seeds and replicated on Climate-FEVER (Diggelmann et al., 2020). Third, we show that retrieval metrics rank the defects in the wrong order of harm, that stale copies matter mainly to verifiers that read each passage closely, and that very small chunks make a fine-tuned verifier confidently wrong.

All code, per-claim predictions, corruption logs and fine-tuned models are publicly available.[1]

[1] https://github.com/Likhithvenkatanalla/rag-dq-benchmark, archived at https://doi.org/10.5281/zenodo.

2 Related Work

Robustness of retrieval-augmented models. Language models are easily distracted by irrelevant context (Shi et al., 2023), make poor use of information in the middle of long inputs (Liu et al., 2024), and can be trained to disregard irrelevant passages (Yoran et al., 2024). The RGB benchmark (Chen et al., 2024) evaluates noise robustness, negative rejection and counterfactual robustness of RAG readers, and Cuconasu et al. (2024) report that adding random documents to the context can even improve accuracy. These studies edit the reader's input directly. We instead edit the collection and let the retriever decide what the reader sees, which is closer to how such problems arise in deployed systems.

Knowledge conflicts. Longpre et al. (2021) substitute entities in passages to create conflicts with a model's parametric knowledge, and Xie et al. (2024) find that LLMs readily accept conflicting evidence when it is coherent. Our stale copies produce a different kind of conflict, one that lives inside the collection: the original document and its reversed copy are both retrievable, much as an outdated and a current version of a guideline can coexist in a document store.

Duplicates and data quality. Near-duplicates are pervasive in web-scale text, and removing them improves language-model training (Lee et al., 2022). Retrieval benchmarks such as BEIR (Thakur et al., 2021) compare retrievers across many clean collections. We take the opposite approach and hold the task fixed while degrading the collection.

Claim verification. FEVER (Thorne et al., 2018), SciFact (Wadden et al., 2020) and Climate-FEVER (Diggelmann et al., 2020) pair claims with evidence and a verdict. This makes the path from evidence to answer measurable, which is why we use them here.

3 Experimental Setup

3.1 Data

SciFact contains expert-written scientific claims and a corpus of 5,183 abstracts. We evaluate on the 300 development claims (124 SUPPORT, 64 CONTRADICT, 112 NEI). Of these, 188 have gold evidence, drawn from 184 distinct abstracts. The 809 training claims, which do not overlap with the development set, are used only for training and for tuning thresholds.

Climate-FEVER pairs real-world climate claims with five annotated Wikipedia sentences each. We build a closed corpus from all 5,240 distinct evidence sentences, which is close in size to SciFact; each sentence becomes one record, titled with the name of its article. After removing the 154 claims labelled DISPUTED, 1,381 claims remain (654 SUPPORT, 253 CONTRADICT, 474 NEI). The gold records of a claim are its evidence sentences whose label matches the claim's verdict, and 907 claims have at least one. No model in this study is trained on Climate-FEVER.

3.2 Pipeline

Retrieval. We use BM25 (Robertson and Zaragoza, 2009) over whitespace tokens and dense retrieval with BGE-small-en-v1.5 (Xiao et al., 2023) and exact cosine search (Johnson et al., 2019). Recall@k is the fraction of a claim's gold records found in the top k, averaged over claims that have evidence.

Verification. Each verifier sees the claim together with the top three records from dense retrieval and outputs SUPPORT, CONTRADICT or NEI. *Qwen-1.5B* and *Qwen-3B* are Qwen2.5-1.5B-Instruct and Qwen2.5-3B-Instruct (Qwen Team, 2024), prompted zero-shot to answer with a single word and decoded greedily. *NLI zero-shot* is the cross-encoder `nli-deberta-v3-xsmall` (He et al., 2023) as released. *NLI fine-tuned* is the same model after three epochs of training on 1,123 pairs built from the SciFact training claims: each gold abstract with its label, plus NEI examples taken from the cited abstracts of claims without evidence and from the highest-ranked non-gold BM25 abstract of each claim with evidence. We train it with three random seeds. The NLI verifiers score every (claim, record) pair separately. A claim receives SUPPORT or CONTRADICT, whichever has the higher maximum probability across its three records, unless neither reaches a threshold, in which case it receives NEI. The threshold maximises macro-F1 on 20% of the training claims held out from training; the development set is never used for tuning.

Reranking. In one experiment a MiniLM cross-encoder (Nogueira and Cho, 2019), either as released or fine-tuned on SciFact training data, re-

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orders the dense top ten before verification.

3.3 Defects

Each defect is applied to 10, 25 or 50% of the gold-evidence records, so that it actually reaches the claims under evaluation. Claims and labels are never modified.

- **Duplicate** adds a lightly edited copy of a record. If the record has more than two sentences, one is dropped; about 5% of the words are deleted, and one pair of adjacent words per sentence is swapped.
- **Stale** adds a copy whose directional findings are reversed by rule (*increased*↔*decreased*, *higher*↔*lower*, "is associated" → "is not associated", and so on). For Climate-FEVER the word list also includes climate terms such as *warming*↔*cooling* and *rise*↔*fall*. When a record contains none of these words, its copy is identical to the original, so we also analyse the reversed copies on their own.
- **Parse** replaces a record with a broken version of it: truncated at 30 to 60% of its length, with its sentences shuffled, or with OCR-style character errors.
- **Chunk** splits every SciFact abstract into records of one, three or five sentences. It does not apply to Climate-FEVER, whose records are already single sentences.

A copy counts as a separate document, so retrieving it does not count as retrieving the gold record. A chunk, by contrast, counts as the abstract it was cut from.

3.4 Statistical analysis

All runs on a dataset are evaluated on the same claims, so we compare them in pairs. For every claim we note whether the answer went from right to wrong (*lost*) or from wrong to right (*gained*) relative to the clean corpus, report both counts, and test them with an exact McNemar test (McNemar, 1947). Because we run many such tests without correcting for multiple comparisons, we only treat an effect as a finding when its direction repeats across corruption seeds and, for the fine-tuned verifier, across training seeds.

Corpus	BM25	Dense	+ rerank
clean	0.615	0.744	0.805
duplicate 10/25/50%	.594/.541/.463	.689/.668/.604	.774/.751/.644
stale 10/25/50%	.583/.530/.466	.691/.622/.551	.752/.657/.571
parse 10/25/50%	.615/.599/.578	.744/.734/.739	.795/.799/.758
chunk 5/3/1 sent.	.583/.590/.520	.749/.742/.762	.803/.818/.784

Table 1: SciFact recall@1 over the 188 claims with evidence (corruption seed 0). Copies leave recall@10 unchanged (0.883 for BM25, 0.957 for dense retrieval).

4 Results on SciFact

4.1 Clean baseline

Dense retrieval outperforms BM25 at every cutoff, with recall@1 of 0.744 against 0.615 and recall@10 of 0.957 against 0.883. Qwen-1.5B reaches 0.557 accuracy and 0.502 macro-F1, comfortably above the majority class (0.413 and 0.195), but it answers SUPPORT for 200 of the 300 claims. The fine-tuned NLI verifier reaches 0.650, 0.680 and 0.667 with its three training seeds, and each seed is significantly better than Qwen-1.5B (lost/gained 32/60, 36/73 and 35/68; $p \leq 0.005$). Without fine-tuning, the same NLI model manages only 0.353. Doubling the size of the LLM closes the gap. Qwen-3B reaches 0.667 accuracy and 0.655 macro-F1, significantly better than Qwen-1.5B (34 lost, 67 gained; $p = 0.001$) and on par with the fine-tuned verifier, without any task-specific training. Its answers are also far more balanced (109 SUPPORT, 96 CONTRADICT, 95 NEI).

4.2 Retrieval

Copies push the gold abstract out of first place, and the more copies there are, the larger the drop (Table 1, Figure 1). Recall@10 does not move at all, because a copy typically lands just above its original. Averaged over five corruption seeds, dense recall@1 at the 50% level falls from 0.744 to 0.584 ± 0.019 with duplicates and to 0.528 ± 0.031 with stale copies. The fine-tuned reranker raises clean recall@1 to 0.805 and wins back part of the loss caused by duplicates, but very little of the loss caused by stale copies. This is to be expected, since a reversed copy is just as relevant to the claim as its original and a relevance model has no reason to prefer one over the other. Broken parsing puts some documents out of BM25's reach (recall@10 drops to 0.845 at 50%) while leaving dense retrieval almost untouched. Chunking hurts BM25 at every size and has no clear effect on dense retrieval.

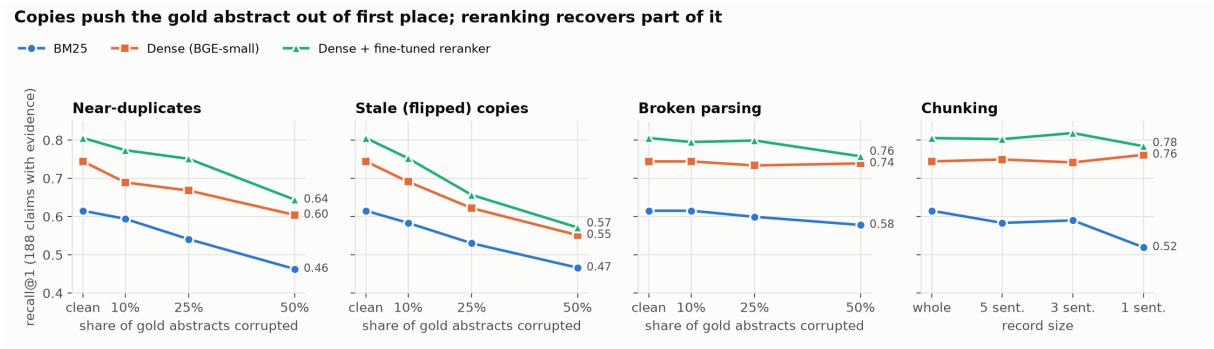


Figure 1: SciFact recall@1 by corruption level for BM25, dense retrieval and dense retrieval followed by the fine-tuned reranker (corruption seed 0).

Setting	Dense R@1	NLI acc.	Lost / gained
clean	0.744	0.650	–
dup. 10%	.704 (.015)	.649 (.003)	3 / 1
dup. 25%	.665 (.026)	.653 (.005)	5 / 10
dup. 50%	.584 (.019)	.657 (.009)	7 / 18
stale 10%	.696 (.004)	.647 (.009)	8 / 3
stale 25%	.624 (.011)	.646 (.010)	16 / 10
stale 50%	.528 (.031)	.649 (.005)	23 / 22
parse 10%	.744 (.004)	.643 (.007)	11 / 1
parse 25%	.736 (.008)	.641 (.006)	18 / 4
parse 50%	.733 (.010)	.631 (.006)	41 / 13

Table 2: SciFact across five corruption seeds (mean and standard deviation): dense recall@1, accuracy of the fine-tuned verifier (training seed 0), and claims lost and gained relative to the clean corpus, summed over seeds.

Corpus	Qwen-1.5B	NLI zero-shot	NLI fine-tuned
clean	0.557	0.353	0.666 (.015)
dup. 10/25/50%	.543/.547/.557	.360/.367/.353	.670/.670/.677
stale 10/25/50%	.553/.550/.560	.353/.357/.347	.672/.661/.656
parse 10/25/50%	.550/.547/.530	.357/.347/.340	.664/.658/.647
chunk 5/3/1	.543/.573/.583	.390/.353/.333	.667/.631/.573

Table 3: SciFact verification accuracy on 300 claims (corruption seed 0). For the fine-tuned verifier we give the mean over three training seeds, with the standard deviation on the clean corpus.

4.3 Verification

For most defects the change in accuracy is small (Table 3, Figure 2), so we look at the paired, claim-level counts instead.

Broken parsing costs correct answers. Across five corruption seeds, the fine-tuned verifier (training seed 0) lost more correct answers than it gained in 14 of the 15 parse runs, and in no run did it gain more than it lost (Table 2). Summed over seeds, it lost 41 answers and gained 13 at the 50% level, and lost 18 and gained 4 at 25%. None of these runs is significant on its own at $p < 0.05$ (the strongest has 12 lost against 4 gained), which is exactly why the repetition matters: if parsing had no effect, a net loss on all five seeds would occur with probability $1/32$ at each rate. The other verifiers point the same way. Qwen-1.5B lost 8 answers and gained none at 50% ($p = 0.008$), all three training seeds of the fine-tuned verifier lost more than they gained on corruption seed 0, and Qwen-3B lost more than it gained at both 25% (5 against 2) and 50% (13 against 7).

Single-sentence chunks hurt the fine-tuned verifier. The effect holds for every training seed (mean accuracy 0.573; lost/gained 58/38, 65/32 and 64/34, with $p = 0.052$, 0.001 and 0.003). The model was trained on whole abstracts, and five-sentence chunks do it no harm. Neither Qwen model is hurt by chunking.

Duplicates leave verification unchanged. Although duplicates have the clearest effect on retrieval, the gold abstract still reaches the top three for almost as many claims as before (87.2% on the clean corpus and 86.7% at 50%), and a near-copy carries the same evidence anyway. Over five seeds the fine-tuned verifier even gained slightly more answers than it lost at 25 and 50% (12 lost, 28 gained in total).

Reranking does little for verification. Verifying from the reranked top three changes accuracy by between 0 and +2.3 points across the 13 corpora, and no change is significant. The gold abstract is usually already among the dense top three, and its exact position there matters little to the verifier.

4.4 Conflicting evidence

To study genuine conflicts, we restrict the analysis to claims whose top three records included a stale copy in which at least one finding was actually reversed, and compare each of these claims with its

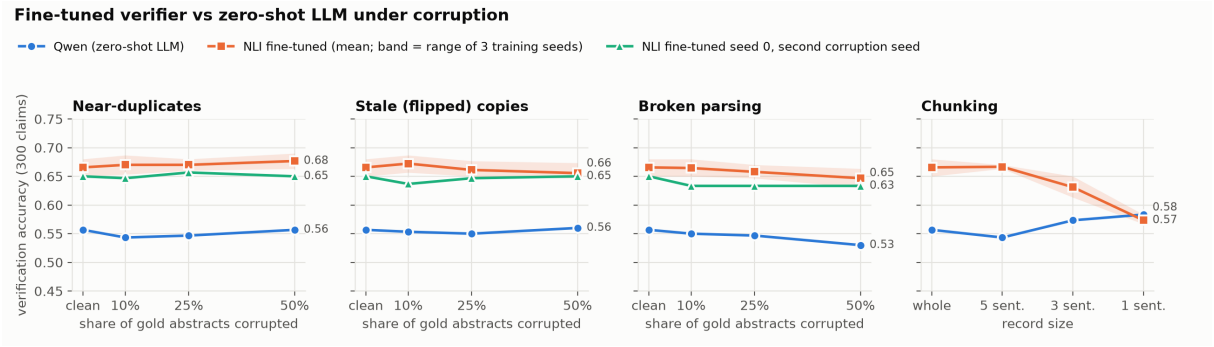


Figure 2: SciFact verification accuracy by corruption level. The orange band spans the three training seeds of the fine-tuned verifier. The green line shows the seed-0 verifier on corruption seed 1, whose own clean accuracy is 0.650.

outcome on the clean corpus. At the 50% level on corruption seed 0, 78 claims meet this condition. The fine-tuned verifier changed its answer on 13, 19 and 16 of them (training seeds 0, 1 and 2), and 11, 15 and 12 of those changes were to CONTRADICT. Qwen-1.5B changed only 6 answers and moved 1 to CONTRADICT, which fits its general preference for SUPPORT. One might suspect that this indifference is simply a symptom of that bias, but Qwen-3B, which is as accurate as the fine-tuned verifier and much more balanced, behaves the same way: it changed 8 of the 78 answers and moved 3 to CONTRADICT (4 lost, 4 gained).

The behaviour of the fine-tuned verifier is stable across corruption seeds. Of the 48 answers it changed at the 50% level over five seeds, 35 moved to CONTRADICT, and claims whose context contained only unreversed copies never changed. Even so, no verifier's accuracy changes significantly on SciFact. The fine-tuned verifier is misled on some claims and corrected on others (23 lost and 22 gained over five seeds), and the reversed copies do not make it confidently wrong more often: on these claims it gave 55 such answers on the clean corpus and 56 with stale copies. The count rose on two seeds and fell on three, so an increase that appeared on a single seed in our earlier analysis does not hold up.

4.5 Calibration

We assign each NLI answer a confidence score. For SUPPORT or CONTRADICT this is the highest probability of that label among the three records, and for NEI it is one minus the highest SUPPORT or CONTRADICT probability. We then compare confidence with correctness. The Qwen models produce no probabilities and are left out of this

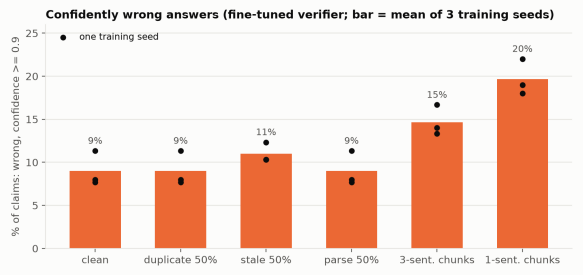


Figure 3: SciFact: share of claims that the fine-tuned verifier answers wrongly with confidence of at least 0.9. Bars show the mean over three training seeds and dots show individual seeds.

analysis. Even on clean data the fine-tuned verifier is overconfident: its mean confidence is 0.75–0.80 while its accuracy is 0.65–0.68, its expected calibration error (ECE; Guo et al., 2017) is 0.14–0.18, and 8–11% of claims are answered wrongly with a confidence of at least 0.9, which we call *confidently wrong*. Copies and broken parses hardly change these figures, but small chunks do. With one-sentence records the verifier becomes more confident while its accuracy drops, its ECE roughly doubles to 0.31–0.32 for all three training seeds, and 18–22% of claims are confidently wrong (Figure 3).

5 Replication on Climate-FEVER

Climate-FEVER is a demanding test for our findings. Its records are single Wikipedia sentences, its claims were collected from the web rather than written by experts, and none of our verifiers was trained on it. Retrieval is much harder than on SciFact: a gold sentence reaches the dense top three for only 53% of claims with evidence, compared with 87% on SciFact. Both verifiers are weak as well. Qwen-1.5B reaches 0.508 accuracy against

	Retrieval		Qwen-1.5B		NLI fine-tuned (SciFact)	
Corpus	Gold in top-3	Dense R@10	Acc.	Lost / gained	Acc. (3 seeds)	Lost / gained at 50% (seeds 0/1/2)
clean	0.534	0.522	0.508	–	0.458	–
duplicate 10 / 25 / 50%	.526 / .517 / .494	.515 / .499 / .486	.511 / .520 / .516	5/9, 4/21, 21/33	.459 / .457 / .456	29/24, 25/21, 24/24
parse 10 / 25 / 50%	.531 / .515 / .504	.509 / .495 / .475	.510 / .511 / .506	3/6, 7/12, 21/19	.457 / .452 / .458	38/32, 27/30, 28/31
stale 10 / 25 / 50%	.528 / .516 / .496	.516 / .502 / .489	.509 / .511 / .508	4/6, 11/15, 26/27	.453 / .446 / .433	55/14, 46/16, 48/15

Table 4: Climate-FEVER, 1,381 claims (corruption seed 0). *Gold in top-3* is the share of claims with evidence whose dense top three contains a gold sentence. *Lost / gained* counts claims that were right on the clean corpus and wrong on the corrupted one, and the reverse.

a majority baseline of 0.474 and predicts NEI for only 89 claims, and the SciFact-trained NLI verifier transfers poorly, reaching 0.458 on average (0.450–0.468) and falling significantly below Qwen-1.5B for every training seed ($p \leq 0.039$). We therefore focus on the effects that do not depend on a verifier being accurate in absolute terms (Table 4).

Stale copies hurt the NLI verifier far more than on SciFact. At the 25 and 50% levels every training seed lost many more correct answers than it gained. At 50% the counts were 55/14, 46/16 and 48/15, all with $p < 0.001$, and accuracy declined steadily with the corruption rate for each seed. Among the 356 claims with a reversed copy in their top three at 50%, the verifier changed between 84 and 101 answers, roughly half of them to CONTRADICT, and became confidently wrong more often under all three seeds (98 to 119, 95 to 120 and 79 to 107). Qwen-1.5B once again hardly reacted (26 lost, 27 gained). We think the difference from SciFact comes from the length of the records. A one-sentence record offers nothing else that could outweigh a reversed finding, whereas a full abstract usually does.

Copies displace gold records without hurting verification. Duplicates and stale copies both reduce the gold-in-top-3 rate by about four points at 50%. Duplicates nonetheless leave accuracy essentially unchanged; the fine-tuned verifier moves by at most 0.4 points, and Qwen-1.5B even improves at 25% (4 lost, 21 gained).

Broken parsing shows no consistent effect. With one-sentence records, a third of the parse corruptions (shuffling the sentence order) change nothing at all, and truncation still leaves part of a short sentence intact. The lost and gained counts point in both directions across verifiers and seeds.

6 Discussion

Retrieval metrics can rank defects in the wrong order. On SciFact, duplicates and stale copies cause the largest drops in recall@1, yet they do not reliably change verification, because the verifier reads the top three records and the gold abstract seldom drops out of them. Broken parsing is the opposite case. Retrieval barely notices it, but it is the defect that most consistently costs correct answers, for both LLMs and for the fine-tuned verifier under all of its training and corruption seeds, because the verifier is handed a truncated or garbled version of the right abstract. A team that tracked retrieval metrics alone would prioritise these defects the wrong way round. Evaluating final answers on a deliberately corrupted copy of one's own collection is inexpensive and avoids this mistake.

Whether conflicting evidence matters depends on the reader. The fine-tuned NLI verifier judges each record on its own and responds to reversed copies. On SciFact it shifts answers towards CONTRADICT without a net loss in accuracy, and on Climate-FEVER, where each record is a single sentence, it loses many correct answers and becomes confidently wrong more often. Both zero-shot LLMs mostly ignore the reversed copies, and this includes Qwen-3B, which is as accurate as the fine-tuned verifier on clean SciFact. Reranking by relevance cannot fix the problem, since a reversed copy is exactly as relevant as its original. Detecting conflicting versions requires other signals, such as dates, provenance, or deduplication before documents are indexed.

Chunk size should match the verifier's training. The fine-tuned verifier, which was trained on whole abstracts, performs worse on single sentences and grows more confident as it does so. Chunking choices should therefore follow the input format a verifier was trained on, and its confidence scores should not be trusted outside that format.

Limitations

On SciFact an effect involving only a few of the 300 claims is within noise, and our McNemar p-values

are not corrected for multiple comparisons. Our conclusions therefore rest on effects that repeat across corruption seeds, training seeds, verifiers and datasets. The Qwen-3B and Climate-FEVER experiments use a single corruption seed. Stale copies are generated by rule-based word substitution and may read less naturally than genuinely outdated documents. Some copies contain nothing to reverse (about 20% on SciFact and 32% on Climate-FEVER), which is why we analyse the reversed copies separately. Because the defects target gold-evidence records, their effect on the evaluated claims is easy to see, but this does not reflect how defects are distributed in real collections. Finally, all models are small (BGE-small, Qwen2.5 up to 3B, DeBERTa-v3-xsmall), since every experiment ran on free CPU hardware. Larger readers may react differently, and we did not study open-ended generation beyond a one-word verdict.

Ethics and Reproducibility Statement

SciFact and Climate-FEVER are public research datasets. We download them from their official releases and do not redistribute them. All experiments ran on free GitHub Actions CPU runners, and the workflows that produced every reported number are included in the repository. Each corrupted corpus is regenerated from its seed and checked against the committed log of changes before use. An AI assistant (Claude) was used to help write code and to draft and edit text; the author reviewed all of it, and every reported number comes from runs whose outputs are published in the repository.

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